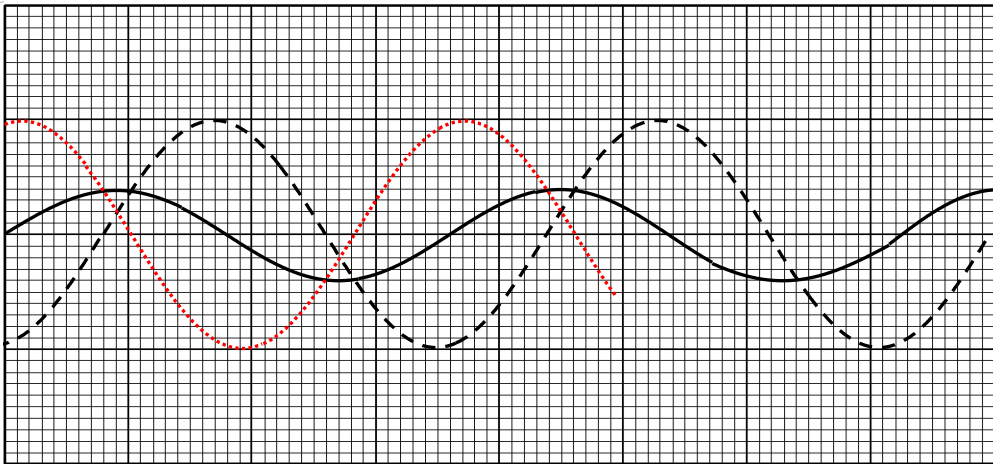


9	(a)	When two or more waves meet at a point at the same time, the resultant displacement is the vector sum of the individual displacements at that point at that time. [2 marks for flawless definition. 1 mark for somewhat correct definition.]	[2]
	(b)		[2]
	(c)	(i) The sound wave reflects off the bench. The incident and reflected waves superpose/interfere with each other. Maximum amplitude where the waves interfere constructively; minimum amplitude where they interfere destructively.	[1] [1] [1]
		(ii) As x increases, the incident wave is stronger and the reflected wave weaker. There is more complete cancellation of the incident and reflected waves at small values of x and less complete cancellation at larger values of x.	[1] [1]
		(iii) Distance between 2 consecutive nodes = $\lambda/2 = (18.0 - 2.6)/3 = 5.13 \text{ cm}$ $\lambda = 10.26 \text{ cm}$ $f = 340/0.1026 = 3310 \text{ Hz}$	[1] [1] [1]
	(d)	(i) An interference pattern is seen on the screen.	[1]
		(ii) $\Delta\phi = \frac{1.4}{1.6} \times 2\pi$$= 5.5 \text{ rad}$	[1] [1]
		(iii) $x = \frac{14.8 - 5.2}{6} = 1.6 \text{ mm}$$a = \frac{\lambda D}{x} = \frac{650 \times 10^{-9} \times 1.5}{1.6 \times 10^{-3}} = 0.61 \text{ mm}$	[1] [1]
		(iv) Distance of 6th order from principal axis = $1.6 \times 6 = 9.6 \text{ mm}$ $\theta = \tan^{-1}\left(\frac{0.0096}{1.5}\right) = 0.0064 \text{ rad}$	[1] [1]

			$b = \frac{1\lambda}{\sin \theta} = \frac{650 \times 10^{-9}}{\sin(0.0064 \text{ rad})} = 0.102 \text{ mm}$	[1]
--	--	--	---	-----